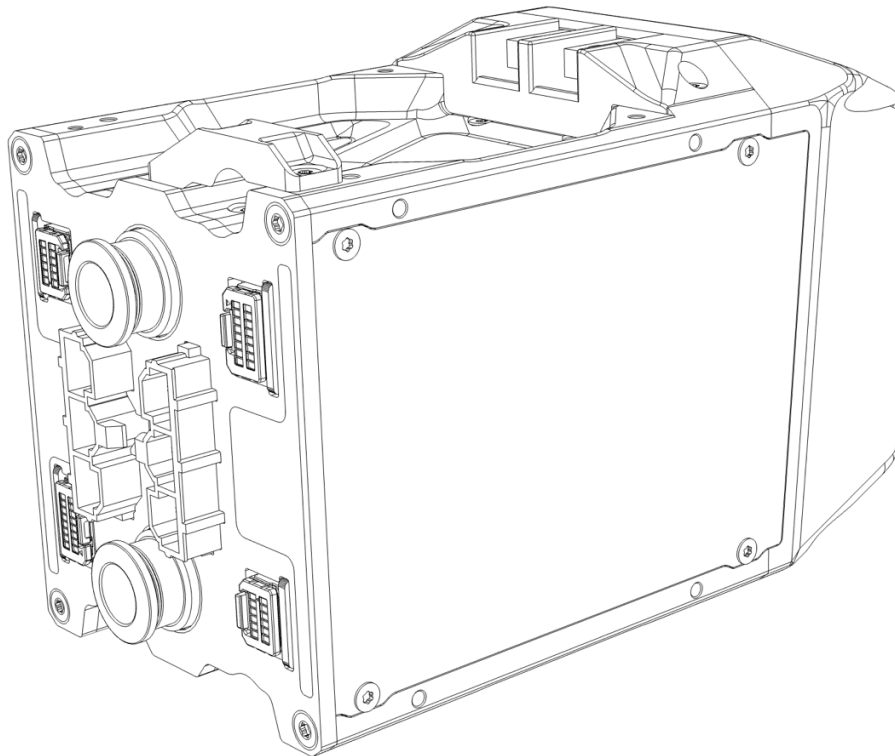


# DTI FORMULA STUDENT INVERTER

## Specification for F-SIC inverter

Version 1.4

**Valid for F-SIC-A inverter**



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# 1. HISTORY AND RELATED DOCUMENTS

## 1.1 Document history

|         |                                                          |
|---------|----------------------------------------------------------|
| 10/2024 | <b>V1.0</b> Document created; basic specifications added |
| 02/2025 | <b>V1.1</b> Related Documents added, release             |
| 05/2025 | <b>V1.2</b> DC characteristics parameter clarifications  |
| 07/2025 | <b>V1.3</b> Added System Efficiency section              |
| 09/2025 | <b>V1.4</b> Thermal parameter corrections                |

## 1.2 Related documents

- DTI CAN manual
- DTI COM manual
- F-SIC-A inverter Application Manual
- F-SIC-A inverter Drawing
- F-SIC-A inverter CAD Model

## 2. TECHNICAL SPECIFICATIONS

### 2.1 DC characteristics

Table 1. DC Characteristics

|                                             | F-SIC-A                  | Unit |
|---------------------------------------------|--------------------------|------|
| Low voltage input range                     | 8 – 20                   | V    |
| Low voltage maximum input current           | 1 / module <sup>1</sup>  | A    |
| Maximum voltage on digital outputs          | 28                       | V    |
| Maximum current on digital outputs          | 100                      | mA   |
| Maximum input voltage on digital inputs     | 28                       | V    |
| Logical LOW voltage on digital inputs       | ≤ 1                      | V    |
| Logical HIGH voltage on digital inputs      | ≥ 2                      | V    |
| High voltage input range                    | 36 – 600                 | V    |
| High voltage absolute maximum input voltage | 620                      | V    |
| High voltage maximum input current          | 55 / module <sup>2</sup> | A    |
| DC link capacitor                           | 75                       | μF   |
| DC link capacitor internal discharge time   | 5                        | Min  |
| Y capacitor <sup>3</sup>                    | 2,2                      | nF   |
| Passive discharge resistor                  | 330                      | KOhm |
| Isolation voltage between LV and HV         | 2500                     | Vdc  |

### 2.2 AC characteristics

Table 2. AC Characteristics

|                                                                 | F-SIC-A (single module) | Unit             |
|-----------------------------------------------------------------|-------------------------|------------------|
| Continuous AC current at <sup>4</sup>                           | 56                      | A <sub>pk</sub>  |
| Maximum AC current for short period (peak current) <sup>5</sup> | 113                     | A <sub>pk</sub>  |
| Maximum AC current for short period (RMS) <sup>6</sup>          | 80                      | A <sub>RMS</sub> |
| Switching frequency                                             | 8 – 30                  | kHz              |
| Maximum power dissipation                                       | 750                     | W                |
| Maximum electric RPM <sup>7</sup>                               | 200 000                 | eRPM             |
| Typical efficiency <sup>8</sup>                                 | 98,15                   | %                |

<sup>1</sup> AWD kit contains 4 modul, so 4 x 1A. FWD kit contains 2 modul, so 2 x 1A.

<sup>2</sup> AWD kit contains 4 modul, so 4 x 55A. FWD kit contains 2 modul, so 2 x 55A.

<sup>3</sup> Y capacitor for common mode noise filtering. Measured between one HV terminal and the house.

<sup>4</sup> This value was measured at a water temperature of 80°C at 10l/min, AC frequency of 300Hz, switching frequency of 30kHz, and a DC voltage of 600Vdc.

<sup>5</sup> The maximum AC current period is depending on the cooling system.

<sup>6</sup> The maximum AC current period is depending on the cooling system.

<sup>7</sup> 200 000 ERPM means 3,33 kHz AC frequency

<sup>8</sup> This value was measured at 550Vdc and 20 kHz switching frequency

### 3. THERMAL AND MECHANICAL SPECIFICATION

*Table 3. Thermal and mechanical specifications*

|                                                       | F-SIC-A              | Unit  |
|-------------------------------------------------------|----------------------|-------|
| Dimensions (h/w/l)                                    | 96,4 / 78 / 153,7    | mm    |
| Weight                                                | 1,1                  | kg    |
| Ambient temperature                                   | -20-85               | °C    |
| Storage temperature                                   | -20-120              | °C    |
| Coolant temperature                                   | -20-120              | °C    |
| IP Protection                                         | IP44                 |       |
| Coolant quantity in device                            | 18                   | mL    |
| Required coolant flow for best continuous current     | 10                   | L/min |
| Required coolant pressure for continuous current      | 170                  | mbar  |
| Max coolant pressure                                  | 1,5                  | bar   |
| Liquid tube connection: Standard 10mm Festo connector | 10                   | mm    |
| Cooling plate material                                | 3D printed aluminium |       |
| Compatible coolant                                    | Distillate water     |       |

For Coolant medium please refer to FS ruling where applicable. Other use-cases should utilize Glycol/Dist. Water mixture with a dilution ratio recommended by the glycol manufacturer and climate conditions.

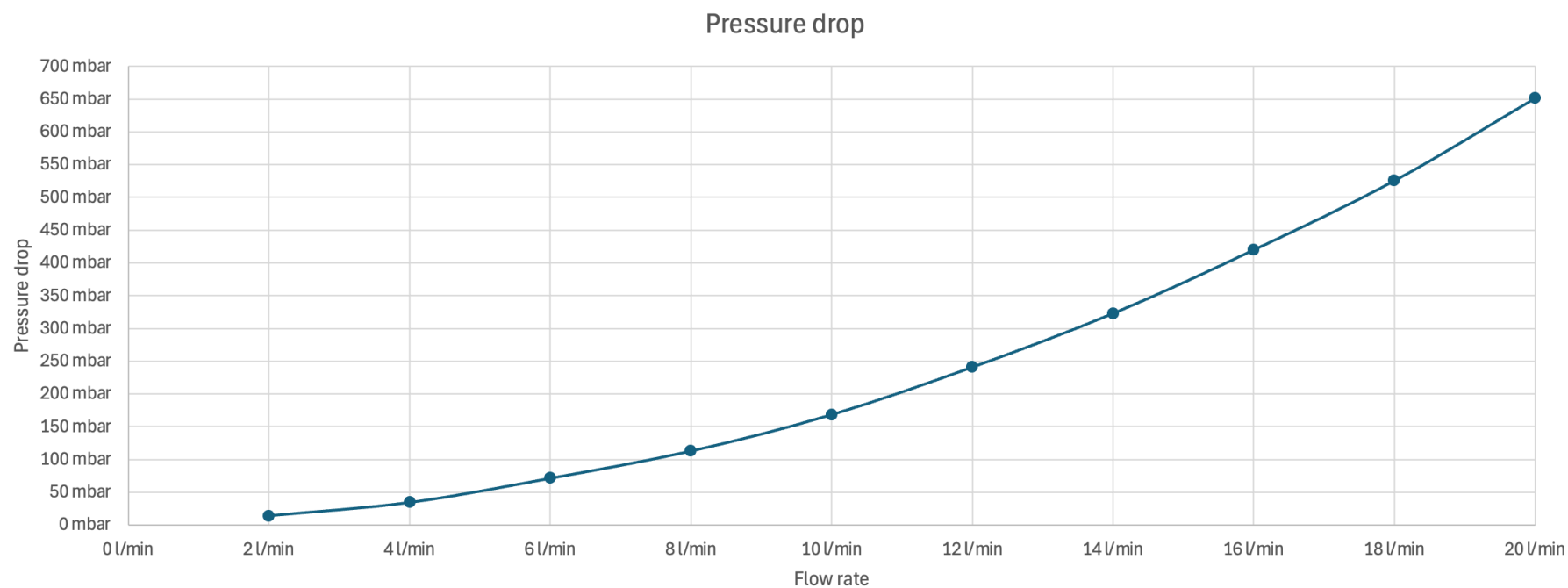


Figure 1 pressure drop between inlet and outlet

## 4. POWER LOSSES

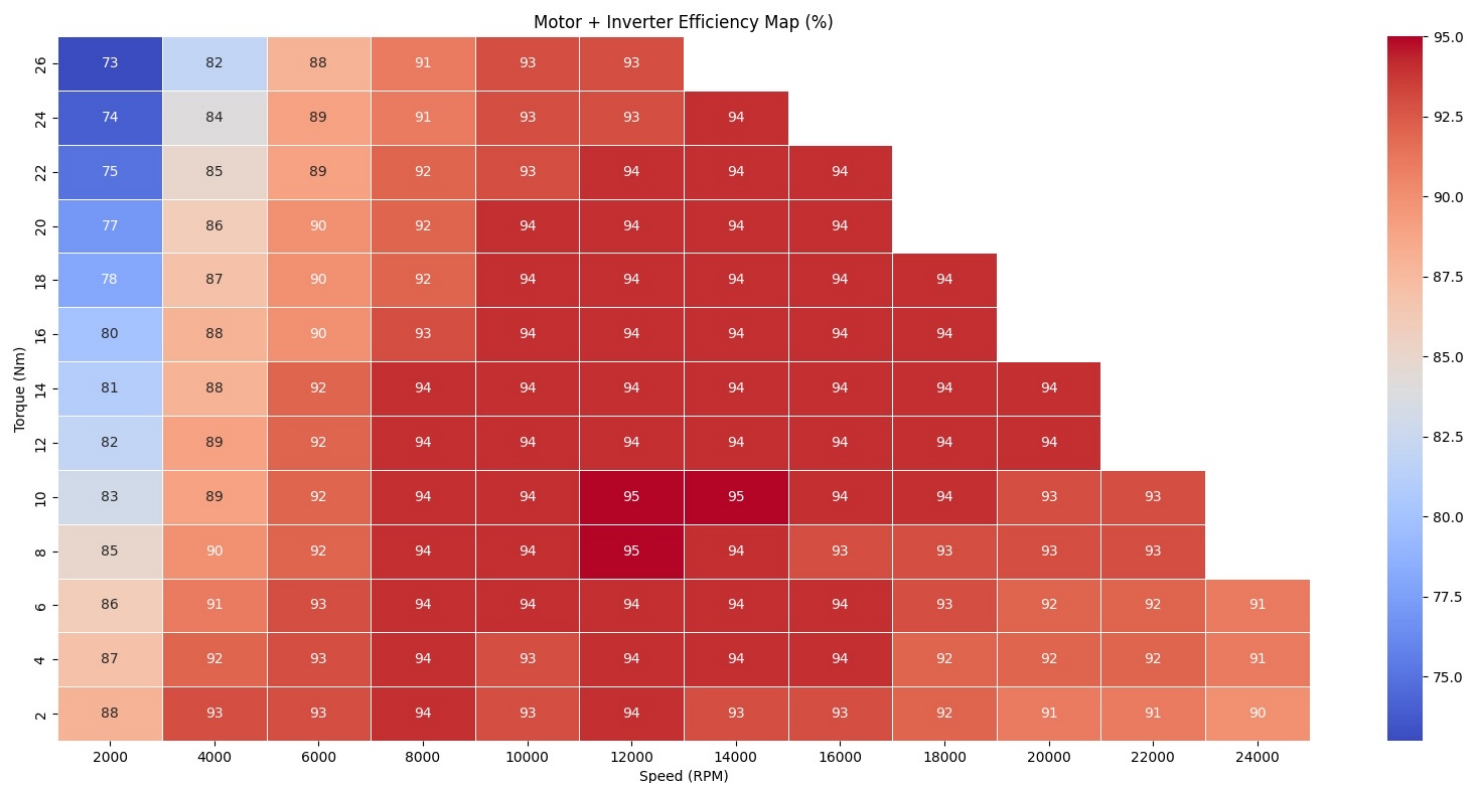
### 4.1 Thermal simulation

We have developed a simulation that considers both thermal performance and the thermal properties of the inverter. This was necessary because calculating the maximum load capacity of an inverter is a complex task. A key aspect of this is the junction temperature, which we continuously monitor to ensure it does not exceed the specified limits. This temperature is influenced by several factors, including:

- DC voltage
- Switching frequency
- AC frequency (motor RPM)
- Case temperature
- AC current

Additionally, the system has a certain thermal capacity, which is associated with a thermal constant. This enhances the transient load capabilities of the inverter. As a result, we created a simulation program that can determine the maximum allowable water temperature and the power dissipation. Detailed instructions for using this simulation can be found in the **Application Manual**.

## 5. SYSTEM EFFICIENCY



## 6. LIST OF TABLES

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